GearDesign Report

Cylindrical gear(pair)

작성일: 2025년 09월 29일

Contents

1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	3.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	0.0000	
Manual normal backlash (mm)	j_bn	0.0000	
Number of teeth	z	28.0000	41.0000
Facewidth (mm)	b	45.0000	45.0000
Profile shift coefficient	x	0.3132	0.2129
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	105.0000	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 (σHlim = 430 Mpa)	DIN 18CrNiMo7 (σHlim = 430 Mpa)
Reference diameter (mm)	d	84.0000	123.0000
Tip diameter (mm)	d_a	91.879 (91.879 / 91.779)	130.277 (130.277 / 130.177)
Root diameter (mm)	d_f	78.379 (78.187 / 78.077)	116.777 (116.585 / 116.475)
Number of teeth spanned	k	4.0000	5.0000
Base tangent length (no backlash) (mm)	W_k	32.817 (32.751 / 32.713)	42.013 (41.947 / 41.910)
Effective Diameter of ball/pin (mm)	D_M	5.5000	5.2500
Diametral measurement over two balls (mm)	M_dk	93.799 (93.652 / 93.567)	131.519 (131.356 / 131.262)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	0.7023		0.7246		1.0000
Bending S.F	0.8932		0.8829		1.4000
Scuffing S.F (Flash Temp.)	3.6689				2.0000
Micropitting S.F	0.4053				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	9999.9000	0.0000	9999.9000	0.0000
Bending Damage [%]	9999.9000	0.0000	9999.9000	0.0000
Contact Life [hr]	43.9000	∞	79.1000	∞
Bending Life [hr]	3.1000	∞	3.7000	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.5335
Overlap Ratio	0.0000
Total Contact Ratio	1.5335
Worst Pitch Line Velocity [m/s]	4.4620
Total Efficiency [%]	99.1509
Worst Mesh Loss [W]	863.2802
Worst Churning Loss [W]	47.5358
Worst Peak-to-Peak T.E. [μm]	15.7414
Gear Total Mass [kg]	2.6096
Hunting tooth	Complete

1.5. Summary - Load cases

	Gear	Time [Hr]	Temperautre [ºC]	Power [kW]	Speed [r/min]	Torque [N.m]	Contact SF (Left)	Bending SF (Left)	Scuffing SF (Left)	Contact SF (Right)	Bending SF (Right)	Scuffing SF (Right)	K_Hβ (Left)	K_Hβ (Right)	Mesh Loss (Left) [kW]	Mesh Loss (Right) [kW]	Churning Loss [kW]	PPTE (Left) [μm]	PPTE (Right) [μm]
LC	Gear1	10000.0000	80.0000	107.2740	1000.0000	1024.3900	0.7023	0.8932	3.6689				1.3000		0.8633		0.0179	15.7414	
1	Gear2			-107.2740	-682.9270	1500.0000	0.7246	0.8829									0.0296		

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	3.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	0.0000	
Manual normal backlash (mm)	j_bn	0.0000	
Number of teeth	z	28.0000	41.0000
Facewidth (mm)	b	45.0000	45.0000
Profile shift coefficient	x	0.3132	0.2129
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	105.0000	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h*_aP	1.0000	1.0000
Dedendum coefficient	h*_fP	1.2500	1.2500
Root radius coefficient	ρ*_fP	0.3800	0.3800
Tip form height coefficient	h*_FaP	0.9667	0.9667
Ramp angle (deg)	α_KP	50.0000	50.0000
Protuberance height coefficient	h*_prP	0.0000	0.0000
Protuberance angle (deg)	α_prP	0.0000	0.0000
Tip alteration (mm)	k.m_n	0.0000	0.0000
Tip radius coefficient (for topping)	ρ*_aP	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 (σHlim = 430 Mpa)	DIN 18CrNiMo7 (σHlim = 430 Mpa)
Young's modulus (N/mm^2)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m^3)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10^-8/°C)	α_coeff	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm^2)	σ_B	1200.0000	1200.0000
Yield strength (N/mm^2)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_ZF	20.0000	20.0000
Mean roughness height, flank (um)	R_ZH	4.8000	4.8000
Roughness average value, root (um)	R_AF	3.0000	3.0000
Roughness average value, flank (um)	R_AH	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm^2)	σ_Flim	430.0000	430.0000
Fatigue strength for contact stress (N/mm^2)	σ_Hlim	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	1.4643	
Transverse module (mm)	m_t	3.0000	
Working normal pressure angle (deg)	α_wn	22.1396	
Transverse pressure angle (deg)	α_t	20.0000	
Working transverse pressure angle (deg)	α_wt	22.1396	
Working transverse pressure angle, e/i (deg)	α_wt, e/i	22.1543 / 22.1248	
Base helix angle (deg)	β_b	0.0000	
Helix angle at operating pitch circle (deg)	β_w	0.0000	
Radial backlash without tolerance (mm)	j_r0	0.0000	
Circumferential backlash without tolerance (mm)	j_t0	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	j_t, e/i	0.0700 / 0.1100	0.0700 / 0.1100
Min/max normal backlash by flank tolerance for each gear (mm)	j_n, e/i	0.0658 / 0.1034	0.0658 / 0.1034
Circumferential backlash, e/i (mm)	j_tw, e/i	0.1331 / 0.2321	
Normal backlash, e/i (mm)	j_nw, e/i	0.1233 / 0.2150	
Angluar backlash, min (deg)	j_θ, min	0.1789	0.1222
Angluar backlash, max (deg)	j_θ, max	0.3122	0.2132
Effective facewidth (mm)	b_eff	45.0000	
Sum of the profile shift coefficients	Sum_x	0.5261	
Generating profile shift coefficient	x_E, e/i	0.2811 / 0.2628	0.1808 / 0.1625
Prevent undercut profile shift coefficient	x_Emin	-0.6377	-1.3981
Max. profile shift coeff. (min top land)	x_Emax	1.2297	1.6118
Reference center distance, x1=x2=0 (mm)	a_d	103.5000	
Working center distance (mm)	a_w	105.0000	
Reference diameter (mm)	d	84.0000	123.0000
Operating pitch diameter (mm)	d_w	85.2174	124.7826
Operating pitch diameter with tolerance (mm)	d_w, e/i	85.2263 / 85.2085	124.7957 / 124.7695
27		91.8792	130.2772
Tip diameter with tolerance (mm)	d_a, e/i	91.8792 / 91.7792	130.2772 / 130.1772

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diamter (mm)	d_f	78.3792	116.7772
Root diameter with tolerance (mm)	d_f, e/i	78.1869 / 78.0770	116.5848 / 116.4749
Tip form diameter (mm)	d_Fa	91.6792	130.0772
Tip form diameter with tolerance (mm)	d_Fa, e/i	91.4572 / 91.3306	129.8643 / 129.7430
Root form diamter (mm)	d_Ff	80.6777	118.9869
Root form diameter with tolerance (mm)	d_Ff, e/i	80.5633 / 80.4996	118.8546 / 118.7802
Active tip diameter (EAP) (mm)	d_Na	91.6792	130.0772
Active tip diameter with tolerance (mm)	d_Na, e/i	91.4572 / 91.3306	129.8643 / 129.7430
Active root diameter (SAP) (mm)	d_Nf	81.2997	120.0673
Active root diameter with tolerance (mm)	d_Nf, e/i	81.4926 / 81.3981	120.2719 / 120.1707
Mean diameter for mass calculation (mm)	d_m	84.9806	123.3785
Gear mass (kg)	m	1.0131	1.5965
Gear inertia (kg.m^2)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.6718	0.6718
Effective tip clearance, e/i (mm)	c, e/i	0.8839 / 0.7570	0.8839 / 0.7570
Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm)	c_NfFf	0.4174	0.6580

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	6.7500	6.7500
Tooth depth, e/i (mm)	h, e/i	6.9011 / 6.7962	6.9011 / 6.7962
Working depth (mm)	h_w	6.0782	
Addendum (mm)	h_a	3.9396	3.6386
Addendum, e/i (mm)	h_a, e/i	3.9396 / 3.8896	3.6386 / 3.5886
Dedendum (mm)	h_f	2.8104	3.1114
Dedendum, e/i (mm)	h_f, e/i	2.9066 / 2.9615	3.2076 / 3.2625
Number of virtual gear teeth	z_n	28.0000	41.0000
Tooth thickness coeff.	s_nc	1.7988	1.7257
Transverse tooth thickness (mm)	s_t	5.3964	5.1772
Normal tooth thickness (mm)	s_n	5.3964	5.1772
Normal tooth thickness at d_a (mm)	s_an	1.9011	2.1504
Normal tooth thickness at d_a, e/i (mm)	s_an, e/i	1.8820 / 1.7808	2.1266 / 2.0339
Normal tooth thickness at d_Fa (mm)	s_Fan	2.0155	2.2507
Normal tooth thickness at d_Fa, e/i (mm)	s_Fan, e/i	2.1357 / 2.0212	2.3423 / 2.2402
Normal space width at d_f (mm)	e_fn	0.0000	2.4074
Normal space width at d_f, e/i (mm)	e_fn, e/i	0.0000 / 0.0000	2.4432 / 2.4647

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	9.4248	
Transverse pitch (mm)	p_t	9.4248	
Normal base pitch (mm)	p_bn	8.8564	
Transverse base pitch (mm)	p_bt	8.8564	
Axial pitch (mm)	p_x	0.0000	
Lead (mm)	p_z	0.0000	0.0000

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_α	13.5812	
Length of path of contact, e/i (mm)	g_α, e/i	13.1587 / 12.8415	
Length of approach POC of pinion (A to C) (mm)	g_f1	6.3233	
Length of approach POC of pinion, e/i (mm)	g_f1, e/i	6.1080 / 6.0733	
Length of recess POC of pinion (C to E) (mm)	g_a1	7.2579	
Length of recess POC of pinion, e/i (mm)	g_a1, e/i	7.0508 / 7.0271	
Length T1 - A (mm)	T1A	9.7344	
T1A tolerance (mm)	e/m/i	10.1296 / 10.0337 / 9.9379	
Length T1 - B (mm)	T1B	14.4592	
T1B tolerance (mm)	e/m/i	14.2402 / 14.1774 / 14.1147	
Length T1 - C (mm)	T1C	16.0577	
T1C tolerance (mm)	e/m/i	16.0695 / 16.0577 / 16.0458	
Length T1 - D (mm)	T1D	18.5908	
T1D tolerance (mm)	e/m/i	18.9860 / 18.8901 / 18.7943	
Length T1 - E (mm)	T1E	23.3156	
T1E tolerance (mm)	e/m/i	23.0966 / 23.0338 / 22.9711	
Length T2 - A (mm)	T2A	29.8363	
T2A tolerance (mm)	e/m/i	29.6037 / 29.5370 / 29.4703	
Length T2 - B (mm)	T2B	25.1115	
T2B tolerance (mm)	e/m/i	25.4852 / 25.3933 / 25.3013	
Length T2 - C (mm)	T2C	23.5130	
T2C tolerance (mm)	e/m/i	23.5304 / 23.5130 / 23.4957	
Length T2 - D (mm)	T2D	20.9799	
T2D tolerance (mm)	e/m/i	20.7473 / 20.6806 / 20.6139	
Length T2 - E (mm)	T2E	16.2551	
T2E tolerance (mm)	e/m/i	16.6289 / 16.5369 / 16.4449	
Length T1 - T2 (mm)	T1T2	39.5707	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	39.5999 / 39.5707 / 39.5415	
Diameter of single contact point A (mm)	d_A	81.2997	130.0772
Diameter of single contact point B (mm)	d_B	84.0647	126.0222
Diameter of single contact point C (mm)	d_C	85.2174	124.7826
Diameter of single contact point D (mm)	d_D	87.2529	122.9629
Diameter of single contact point E (mm)	d_E	91.6792	120.0673
Transverse contact ratio	ε_α	1.5335	
Overlap ratio	ε_β	0.0000	
Total contact ratio	ε_Υ	1.5335	
Total contact ratio with tolerance	e/m/i	1.486 / 1.468 / 1.450	
Specific sliding at Tip	ζ_a	0.5239	0.5223
Specific sliding at Root	ζ_f	-1.0932	-1.1003

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	9.0000	9.0000
Base circle pitch deviation (μm)	f_pb	8.5000	8.5000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	20.0000	19.0000
Profile form deviation (μm)	f_fα	9.5000	9.5000
Profile slope deviation (μm)	f_Hα	7.5000	7.5000
Total profile deviation (μm)	F_α	12.0000	12.0000
Helix form deviation (μm)	f_fβ	11.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.5000	10.0000
Total helix deviation (μm)	F_β	15.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	27.0000	29.0000
Runout (μm)	F_r	24.0000	26.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.5000	8.5000
Single flank composite, total (μm)	F_isT	35.5000	37.5000
Tooth-to-tooth radial composite deviation (μm)	f_idT	14.0000	14.0000
Total radial composite deviation (μm)	F_idT	36.0000	36.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	4.0000	5.0000
Base tangent length (mm)	W_k	32.8166	42.0133
Base tangent length with tolerance (mm)	W_k, e/i	32.7508 / 32.7132	41.9475 / 41.9099
Tolerance of W_k (mm)	ΔW_k, e/i	-0.0658 / -0.1034	-0.0658 / -0.1034
Diameter of contact point (mm)	d_M	85.4517	122.9523
Theoretical diameter of ball/pin (mm)	D_M.theory	5.4396	5.2205
Effective Diameter of ball/pin (mm)	D_M.eff	5.5000	5.2500
Radial single ball measurement (mm)	M_rk	46.8996	65.8060
Radial single ball measurement with tolerance (mm)	M_rk, e/i	46.8260 / 46.7837	65.7243 / 65.6774
Diametral two ball measurement (mm)	M_dk	93.7992	131.5193
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	93.6520 / 93.5674	131.3561 / 131.2623
Tolerance of M_dk (mm)	ΔM_dk	-0.1473 / -0.2319	-0.1632 / -0.2569

3. Detail rating

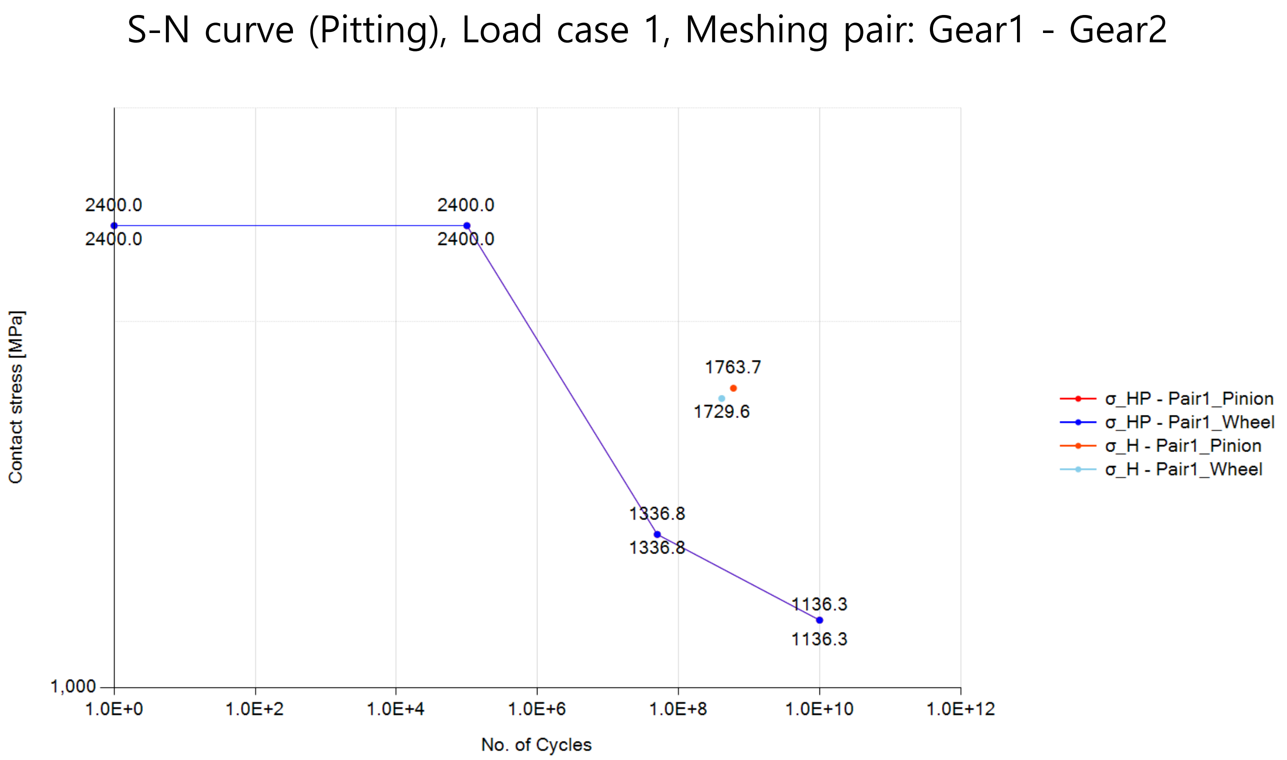
3.1. General

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	1000.0000	682.9270
Nominal tangential load at d (N)	F_t	24390.2440	
Nominal axial load at d (N)	F_a	0.0000	
Nominal radial load at d (N)	F_r	8877.3230	
Nominal normal load at d (N)	F_n	25955.5550	
Nominal tangential load per unit facewidth (N/mm)	w	542.0050	
Circumferential speed at d (m/s)	v	4.3980	
Sliding velocity at Tip dia. (m/s)	v_g	1.2790	1.1140
Nominal tangential load at d_w (N)	F_tw	24041.8140	
Nominal axial load at d_w (N)	F_aw	0.0000	
Nominal radial load at d_w (N)	F_rw	9781.7190	
Circumferential speed at d_w (m/s)	v_w	4.4620	
Running-in value (um)	y_p	0.6370	
Running-in value (um)	y_f	0.7130	
Effective single pitch & profile deviation (um)	f_pb_eff	7.8620	
Effective single pitch & profile deviation (um)	f_fa_eff	8.7870	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_st	1.0000	
Theoretical single stiffness (N/mm/um)	c'_th	18.2760	
Single tooth stiffness (N/mm/um)	c'	12.8010	
Mesh stiffness (for Kv, KHα, KFα), (N/mm/um)	c_yα	17.9230	
Mesh stiffness (for KHβ, KFβ), (N/mm/um)	c_yβ	15.2350	
Mean diameter for K_v (mm)	d_m (K_V)	85.1290	123.5270
Moment of inertia for K_v (kg.mm^2)	J (K_V)	1627.2220	6092.6950
Moment of gear inertia per unit faccewidth (kg.mm)	J*	36.1600	135.3930
Relative gear mass per unit facewidth (kg/mm)	m*	0.0232	0.0405
Relative gear mass per unit facewidth (kg/mm)	m_red	0.0148	
Resonance running speed (rpm)	n_E1	11883.7990	
Resonance ratio	N	0.0840	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_V	1.0280	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	K_Hβ	1.3000	
Face load factor - root	K_Fβ	1.2500	
Transverse load factor - flank	K_Hα	1.0000	
Transverse load factor - root	K_Fα	1.0000	
Number of load cycles	N_L	6.0000×10^8	4.0980×10^8

3.2. Pitting



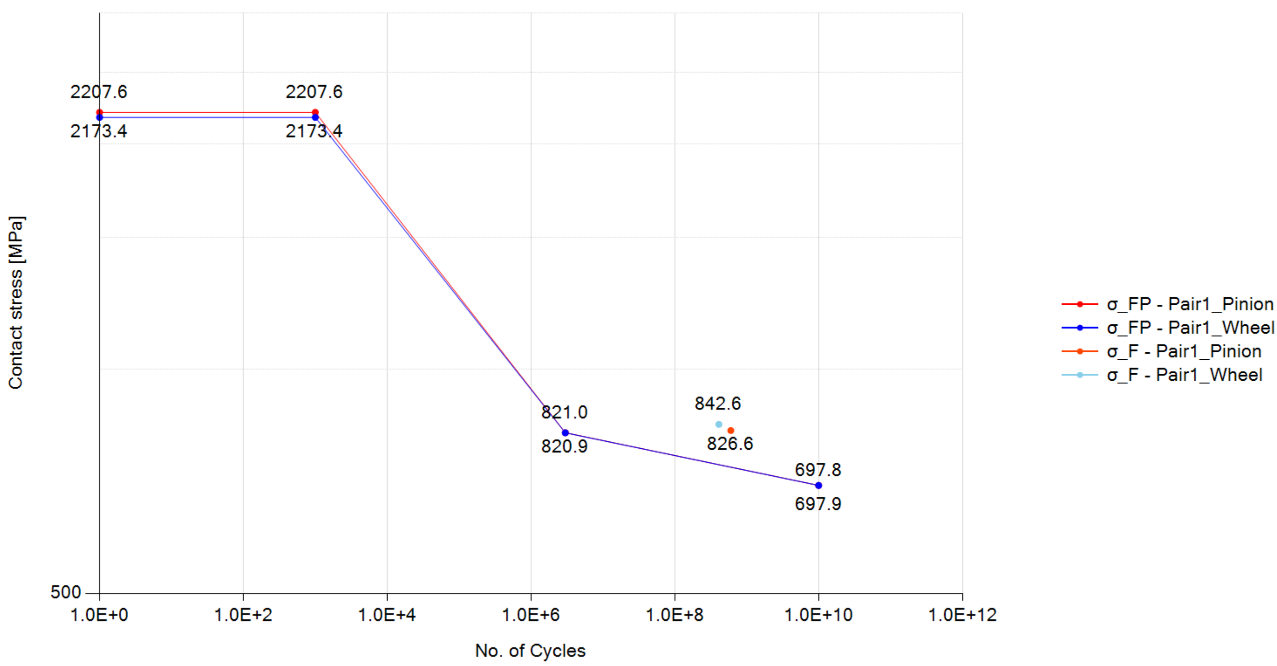
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_ε	0.9070	
Zone factor	Z_H	2.3590	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm^2)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_β	1.0000	
Single pair tooth contact factor (pinion)	Z_B	1.0200	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm^2)	σ_H0	1338.1470	
Contact stress (N/mm^2)	σ_H	1763.7470	1729.6140
Life factor	Z_NT	0.9270	0.9380

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	0.9800	0.9800
Roughness factor	Z_R	0.9620	0.9620
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm^2)	σ_HP_static	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm^2)	σ_HP_mid	0.0000	0.0000
Permissible contact stress - reference (N/mm^2)	σ_HP_ref	1336.7950	1336.7950
Permissible contact stress - long (N/mm^2)	σ_HP_long	1136.2760	1136.2760
Permissible contact stress (N/mm^2)	σ_HP	1238.6900	1253.2650
Safety factor for Contact	S_H	0.7020	0.7250
Contact damage (%)	-	22776.8430	12639.7040
Cycle to failure	-	2.6340×10^6	3.2420×10^6

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.3820	1.4340
Stress correction factor	Y_S	2.0610	2.0250

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_Fen	22.5290	21.6820
Bending moment arm (mm)	h_Fe	3.2800	3.4700
Tooth thickness at root (mm)	s_Fn	6.4800	6.5630
Tooth root radius (mm)	ρ_F	1.4220	1.4500
Helix angle factor	Y_β	1.0000	
Deep tooth factor	Y_DT	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm^2)	σ_F0	514.5480	524.4740
Tooth root stress (N/mm^2)	σ_F	826.6410	842.5880
Life factor	Y_NT	0.8990	0.9060
Relative notch sensitive factor	Y_δrelT	0.9980	0.9980
Surface factor	Y_RrelT	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_ST	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm^2)	σ_FP_static	2207.6110	2173.4440
Permissible tooth root stress - reference (N/mm^2)	σ_FP_ref	821.0430	820.9200
Permissible tooth root stress - long (N/mm^2)	σ_FP_long	697.8860	697.7820
Permissible tooth root stress (N/mm^2)	σ_FP	738.3540	743.9060
Safety factor for Bending	S_F	0.8930	0.8830
Bending damage (%)	-	3.2194×10^5	2.6920×10^5
Cycle to failure	-	1.8640×10^5	1.5220×10^5

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_oil	80.0000	
Lubricant viscosity (mPa.s)	η_oil	13.1510	
Load stage according to FZG A/8, 3/90	S_FZG	12.0000	
Gear density (kg/m^3)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_eff	38.2500	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	905.5120	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	52.9250	
Angle factorr	X_αβ	1.0090	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0800	
Maximum flash temperature (deg)	θ_fl_max	57.3660	
Mean flash temperature (deg)	θ_fl_mean	35.9370	
Total mass temperature (deg)	θ_M	100.2690	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	157.6350	
Safety factor for flash temperature	S_B	3.6690	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.0513
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.0855
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	0.4053
transverse tangential load at reference cylinder per mesh (N)	F_t	24390.2440
transverse load in plane of action (N)	F_bt	25955.5550
reduced modulus of elasticity (N/mm^2)	E_r	2.2637×10^5
thermal contact coefficient of pinion (N/(ms^0.5*K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms^0.5*K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90º)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1020
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm^2)^0.5	Z_E	189.8120
helical load factor	K_Bγ	1.0000
load losses factor	H_v	0.1220
mean coefficient of friction	μ_m	0.0860
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.3330
buttressing factor	X_but,A	0.0000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	3.1530
sliding velocity at point A (m/s)	v_g,A	-1.1140
tangential velocity on pinion at point A (m/s)	v_r1,A	1.0190
tangential velocity on wheel at point A (m/s)	v_r2,A	2.1340
oil inlet/sump temperature (ºC)	θ_oil	80.0000
bluk temperature (ºC)	θ_M	98.3040
flash temperature at point A (ºC)	θ_fl,A	56.7340
contact temperature at point A (ºC)	θ_B,A	155.0380
pressure-viscosity coefficient at 38 ºC (m^2/N)	α_38	1.8390×10^-8
pressure-viscosity coefficient at bulk temperature (m^2/N)	α_θM	1.3440×10^-8
pressure-viscosity coefficient at point A contact temperature (m^2/N)	α_θB,A	1.0050×10^-8
kinematic viscosity at 40 ºC (mm^2/s)	ν_40	68.0000
kinematic viscosity at 100 ºC (mm^2/s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm^2/s)	ν_θB,A	3.2450
dynamic viscosity at 38 ºC (Ns/m^2)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m^2)	η_θM	0.0080
dynamic viscosity at oil inlet/sump temperature (Ns/m^2)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m^2)	η_θB,A	0.0030
normal radius of relative curvature at point A (mm)	ρ_n,A	7.3400
material parameter	G_M	3041.4620
local velocity parameter at point A	U_A	7.1850×10^-12
local load parameter at point A	W_A	0.0002
local sliding parameter at point A	S_GF,A	0.2520
local nominal Hertzian contact stress at point A (Method B) (N/mm^2)	p_H,A	971.4730
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm^2)	p_dyn,A	1255.6720

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	107.2739
Gear ratio	u	1.4643
Tangential speed at working pitch dia. (m/s)	v_wt	4.4620
Tangential force at pitch dia. (N)	F_t	24390.2439
Tooth normal force, transverse section (N)	F_bt	25955.5554
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_Σ	3.3631
Equivalent radius of curvature (mm)	ρ	9.5415
Kinematic viscosity of oil at operating temp. (cSt)	η_oil	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	576.7901
Mean coefficient of friction of the gear mesh	μ_mz	0.0658
Tooth loss factor	H_v	0.1223
Mesh power loss (W)	P_VZP	863.2802
Mesh efficiency (%)	η_z	99.1953

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	1000.0000	682.9268
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immerged depth (mm)	h_e	42.9396	62.1386
Oil density (kg/m^3)	ρ	885.0000	885.0000
Operating Viscocity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	13616.7852	13457.1189
Froude number	Fr	48.0006	32.3965
Centrifugal acceleration (m/s^2)	γ	196.9947	103.9212

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	91.8792	130.2772
Pitch diameter of gear (mm)	d_p	85.8792	124.2772
Facewidth (mm)	b	45.0000	45.0000
Dimensionless drag torque	C_m	0.0213	0.0223
Immersion area of the gear (m^2)	S_m	0.0209	0.0342
Total oil churning loss (W)	P	17.9142	29.6216
Total oil churning torque (N.m)	T	0.1711	0.4142

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000